

A6

October 22, 2025

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[6]: import numpy as np
      from collections import Counter
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[7]: def euclidean_distance(point1, point2):
      return np.sqrt(np.sum((np.array(point1) - np.array(point2))**2))
```

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[8]: def knn_predict(trdata, trlabels, tepoint, k):
      distances = []
      for i in range(len(trdata)):
          dist = euclidean_distance(tepoint, trdata[i])
          distances.append((dist, trlabels[i]))
      distances.sort(key=lambda x: x[0])
      k_nearest_labels = [label for _, label in distances[:k]]
      return Counter(k_nearest_labels).most_common(1)[0][0]
```

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[10]: trdata = [[4, 21], [5, 19], [10, 24], [4, 17], [3, 16], [11, 25], [14, 24], [8, 22],
      ↪          [10, 21], [12, 21]]
      trlabels = ['0', '0', '1', '0', '0', '1', '1', '0', '1', '1']
      tepoint = [8, 21]
      k = 1

      prediction = knn_predict(trdata, trlabels, tepoint, k)
      print(prediction)
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[11]: tepoint = [14, 25]
      k = 1

      prediction = knn_predict(trdata, trlabels, tepoint, k)
      print(prediction)
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[12]: tepoint = [11, 22]
      k = 4
```

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prediction = knn_predict(trdata, trlabels, tepoint, k)
print(prediction)
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[13]: tepoint = [5, 20]
      k = 4

      prediction = knn_predict(trdata, trlabels, tepoint, k)
      print(prediction)
```

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